Computers & Education 252 (2026) 105658

Contents lists available at ScienceDirect

Computers & Education

journal homepage: www.elsevier.com/locate/compedu

Exploring the role of conceptual knowledge and representational competence in an augmented reality physics environment

Kozlova Zoya [a,*], Edelsbrunner Peter A. [a], Warkentin Max [b], Hoyer Christoph [b], Küchemann Stefan [b], Kuhn Jochen [b], Hofer Sarah I. [a]

[a] Department of Psychology, Ludwig-Maximilians-Universität München, Munich, Germany
[b] Chair of Didactics of Physics, Ludwig-Maximilians-Universität München, Munich, Germany

ARTICLE INFO

Keywords:
Augmented reality
Prior knowledge
Individual differences
Physics education

ABSTRACT

Prior knowledge plays a crucial role in learning in STEM education. The potential interplay between components of the prior knowledge, such as representational competence and conceptual understanding, remains remarkably underexplored. Even less is known about how these relationships unfold in augmented reality (AR)-supported learning environments, which offer unique affordances, such as visualizing the invisible, that may shape the role of prior knowledge in new ways.

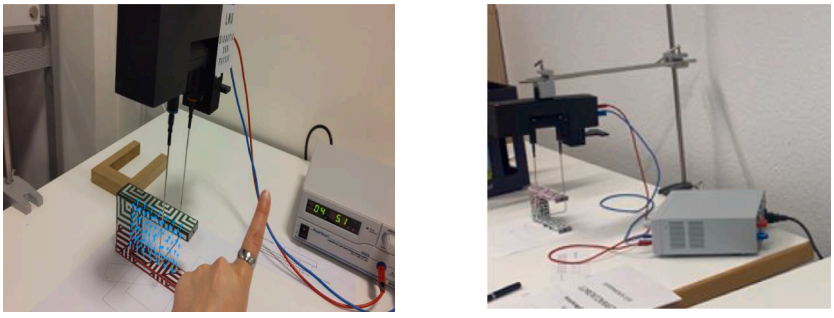
Using a sample of $N = 203$ high school students, we examined how prior knowledge of physics concepts and representations influenced subsequent knowledge of both concepts and representations in AR-supported and traditional learning environments. Learners in the AR condition acquired more conceptual knowledge than those in the traditional condition. A cross-lagged path analysis revealed a consistent, unidirectional relationship from prior representational competence to subsequent conceptual knowledge, when controlling for prior conceptual knowledge for both instructional conditions. These results indicate that representational competence is a robust predictor of conceptual learning, independently of the learning environment.

What role does previously gained knowledge play in the acquisition of new knowledge across domains? To what extent does this relationship depend on the learning environment? These longstanding questions lie at the heart of understanding individual learning mechanisms.

Prior research has consistently demonstrated that existing knowledge serves as a crucial foundation for future learning, guiding the integration of new information (Alexander & Murphy, 1999; Bittermann et al., 2023; Carey, 2000). Domain-specific prior knowledge is a significant determinant of learning outcomes (Alexander & Judy, 1988; Witherby & Carpenter, 2022). This relationship holds across different types of knowledge - declarative, procedural, or mixed - as well as across a range of subject domains and educational levels (Simonsmeier et al., 2022). Simonsmeier et al. (2022) confirmed in a meta-analysis that prior knowledge is strongly correlated with posttest performance ($r = 0.53$), and this effect was consistent even when controlling for intelligence or study design. Recent work by Schneider and Simonsmeier (2025) highlights that it is not just the presence of prior knowledge that matters. In other words, it is the characteristics of this knowledge, such as its accuracy, coherence, and relevance to the learning task, that determine whether it facilitates or hinders learning. However, such findings primarily stem from studies conducted in traditional learning environments, such

* Corresponding author. Leopoldstr. 13, 80802, München, Germany.
E-mail address: zoya.kozlova@psy.lmu.de (K. Zoya).

https://doi.org/10.1016/j.compedu.2026.105658
Received 28 July 2025; Received in revised form 13 April 2026; Accepted 1 May 2026
Available online 14 May 2026
0360-1315/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).



a) AR-assisted experimental set-up

b) Control group experimental set-up

Fig. 2. Experimental set-up.

3.4. Study design

Although the study examined learning outcomes in both the AR and traditional settings, it should not be regarded as a mere media comparison study. Rather, the design was grounded in a theoretically informed approach that controlled for key instructional variables, ensuring that any observed differences could be attributed to the specific affordances of the AR-based visualization rather than to variations in instructional design or learning sequence. Consistent with the considerations outlined by Lawson et al. (2024), the experimental conditions were equivalent in all respects except for the integration of the visual representation of vector fields and field lines. Furthermore, to account for the complexity of the AR-supported environment, we employed the framework developed by Kozlova et al., 2025, which underscores the necessity to equally consider individual differences, learning outcomes, AR affordances, and instructional design when designing a study on the effects of immersive technology (Fig. 3).

We utilized the "visualizing the invisible" and 3D perspective affordance of *AR* to first, make the normally invisible vector fields and field lines visible by digitally superimposing them between the two poles of a real horseshoe magnet and second, add a three-dimensional perspective to the representations. As for the *instructional design*, we used a sequence of six guided experiments that the students had to do independently (i.e., without teacher support, but with guiding working materials & receiving instructions as described above) either in the AR experimental set-up, or in the traditional one. Prior to the experimentation, the students were shown a learning video that aimed to activate their knowledge of representations (e.g., vector fields and field lines). Throughout the experimentation phase, the participants were either supported by the augmented representations in the AR group (Fig. 4a) or by the representations visualized on the screen (screenshots from the previously shown learning video in the control group; Fig. 4b).

In terms of the *learning outcome*, the students were to transfer their knowledge of representations and concepts in the electromagnetism topic to establish the rule for the occurrence and direction of the Lorentz force. The knowledge complexity they were exposed to was high, since the students had to make use of multiple knowledge chunks to grasp new concepts (see Table 1).

In this study, we refer to the type of prior knowledge under investigation as *conceptual knowledge*, which can be defined as an integrated understanding of principles, relationships, and structures that connect facts and procedures within a domain. It enables learners to explain how and why concepts are related, draw inferences, solve novel problems, and transfer understanding to new contexts. This form of knowledge goes beyond knowing isolated facts or how to carry out tasks (procedural knowledge; Anderson et al.,

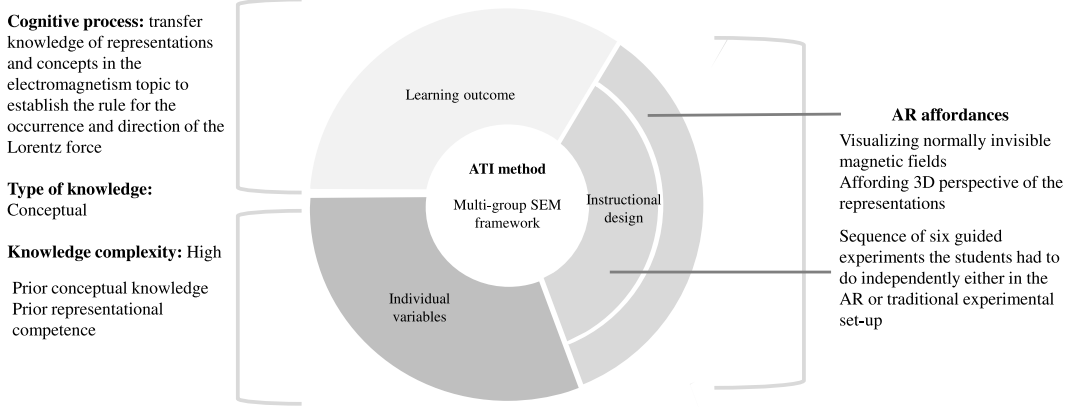


Fig. 3. Learning in Technology-Afforded Environments Triad: a study design framework for AR environments (adapted from Kozlova et al., 2025).

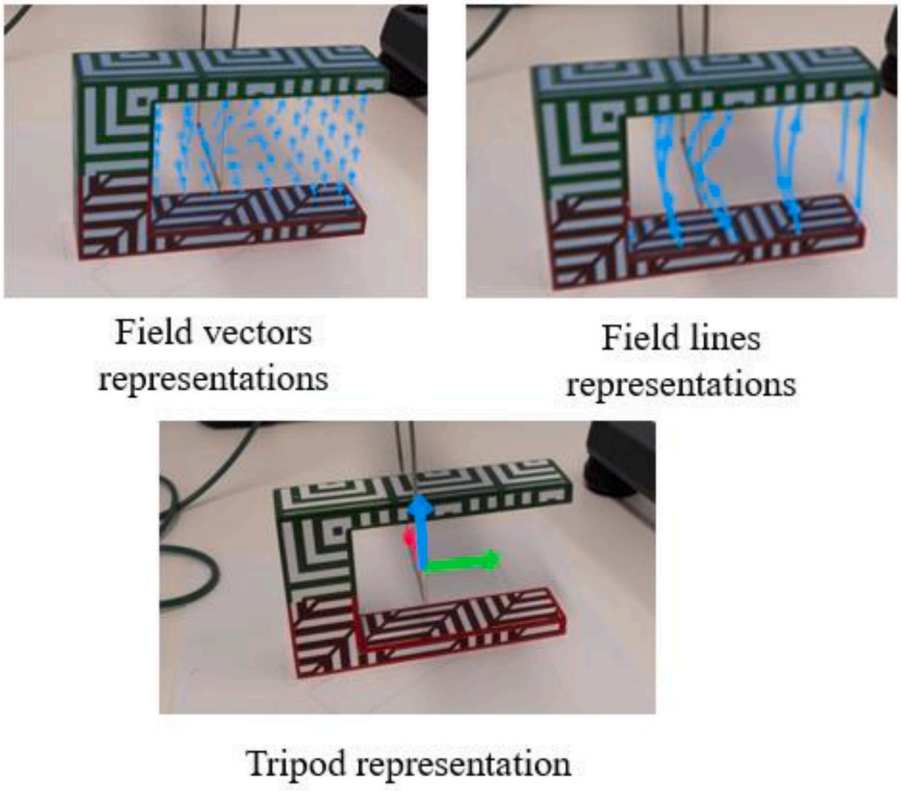


Fig. 4a. Augmented Representations in the Experimentation Phase (AR group).

2004; Baxter et al., 1996; Engelbrecht et al., 2005).

Given the previously identified relationship between representational competence and conceptual knowledge, we also integrated representational competence into the present study. Alongside conceptual knowledge, we treated representational competence as an individual characteristic (prior to the instruction) that may affect the acquisition of both learning outcomes, and as a learning outcome itself (post-instruction).

Rather than incorporating explicit interaction terms, as recommended in the BLINDED Framework (Kozlova et al., 2025), we evaluated the moderation effects of group through a multi-group SEM framework, as explained in more detail in Section 3.5. This procedure is mathematically equivalent but allows higher flexibility, because all model parameters, not just the focal interaction parameters, can be compared between conditions in this approach. By comparing constrained and unconstrained models (specifically prior representational competence and subsequent conceptual knowledge path) across instructional conditions (groups), we tested whether the cross-lagged and autoregressive effects varied significantly between groups.

3.5. Measures

Prior and subsequent representational competence were measured using the Representational Competence of Fields Inventory (RCFI), developed by Küchemann et al. (2021). The RCFI is a diagnostic assessment tool designed to measure students' representational competence in interpreting and translating between vector-field plots and field-line representations in electromagnetism. It consists of 10 items in three core areas: interpreting single representations (e.g., understanding direction or magnitude in vector plots or field lines), translating between vector-field plots and field-line diagrams, and integrating multiple field representations (such as combining vector fields through superposition). The test is content-free by design, meaning the items are not situated within specific topics such as magnetism. This allowed for the prevention of topical confounding of the representational competence test with the instructional topic of electromagnetism. The test nonetheless incorporates concepts, such as vector fields and field lines, that are closely related to the domain of the learning sequence. This connection provided a meaningful foundation for students to activate or acquire representational knowledge that is essential for understanding the Lorentz force, which was addressed later in the intervention for both groups.

Prior and subsequent conceptual knowledge were assessed with an instrument adapted from Edelsbrunner et al. (2023). The prior and subsequent conceptual knowledge assessments mostly used different items. The overlapping items were nine items designed to capture prerequisite fundamental knowledge related to magnetic fields. Namely, it focused on the students' ability to reason about the direction of the magnetic field generated by an electric current in a conductor, the influence of current direction on the magnetic field

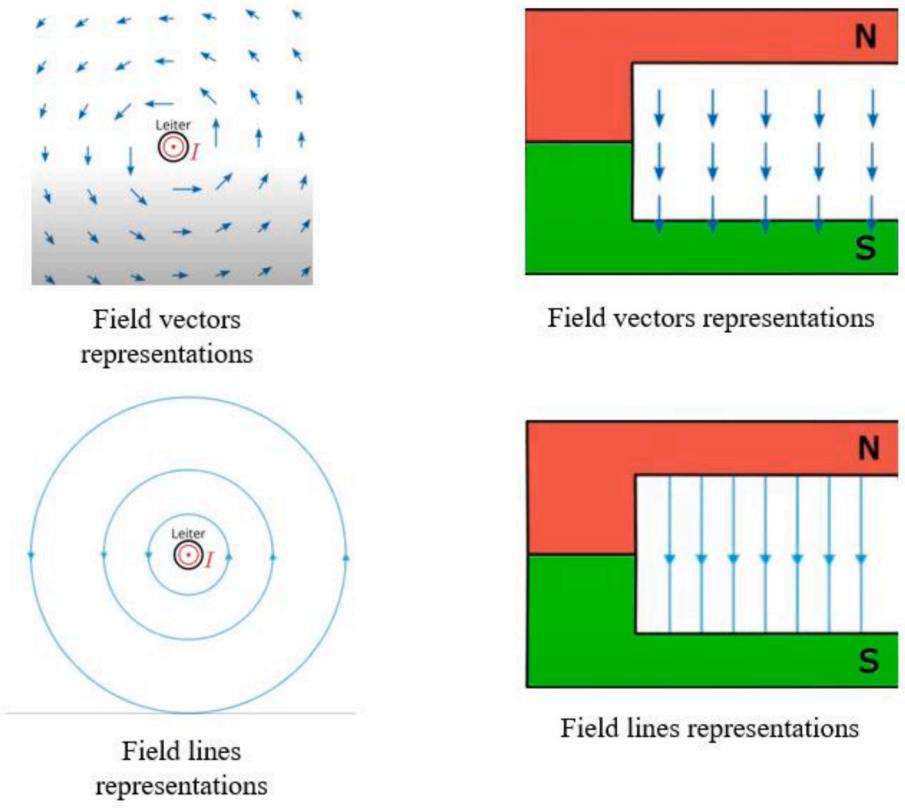


Fig. 4b. Digital Representations in the Experimentation Phase (Control group).

orientation, and the superposition of magnetic fields when multiple sources interact. The pretest did not cover the actual learning goals of our instructional unit. The conceptual knowledge post-test consisted of 23 items, nine of which were identical to the pre-test items. As these nine items were used to assess prerequisite fundamental knowledge and were not addressed by the instructional unit, we did not intend to compare test performance on these items as a pre-post comparison.

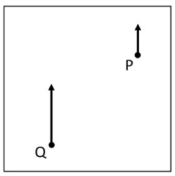
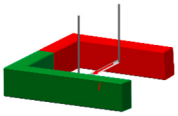
The remaining 14 items of the subsequent conceptual knowledge test assessed students' ability to apply their understanding of the Lorentz force to representations that differed slightly from those used during the intervention. Specifically, students were asked to use the right-hand rule to determine and draw the direction of the Lorentz force acting on one of two parallel current-carrying conductors, based on the magnetic field generated by the other. The content covered in the tests, along with the McDonald's omega reliability coefficients, is presented in Table 1. Although the McDonald's omega values for the prior conceptual knowledge test were somewhat lower than commonly recommended thresholds, we are aware of this limitation. However, we consider this acceptable given the broad and heterogeneous nature of the construct being assessed (Edelsbrunner et al., 2025). The test was intentionally designed to cover a wide range of knowledge components across different subdomains, which likely reduced internal consistency while still reflecting the complexity of the underlying construct (Edelsbrunner et al., 2025; Stadler et al., 2021; Taber, 2018). In addition, based on the guidelines for expected internal consistency reliability reported by Edelsbrunner et al. (2025), a knowledge test with nine items should achieve a value of at least 0.18, which our test easily surpasses.

All test items were scored dichotomously, where a value of 1 was assigned to correct responses and 0 to incorrect or missing responses. For each participant, composite scores were calculated by computing the mean of all item scores within each scale (e.g., prior conceptual knowledge or prior representational competence). As such, resulting scores range from 0 to 1, where 0 indicates no correct answers and one indicates all answers were correct. Intermediate values (e.g., 0.47) represent the proportion of correctly answered items.

3.6. Statistical analyses

All statistical analyses were performed using R Studio (Version 2025.12.1) and relevant statistical packages (e.g., “psych” and “dplyr”; R Core Team, 2026). To assess the difference in the subsequent conceptual knowledge in the AR and control groups, an ANCOVA was conducted with post-test conceptual knowledge as the dependent variable, group (AR vs. Control) as the independent variable, and prior conceptual knowledge and representational competence as covariates. We also controlled for class differences via fixed effects to adjust for the hierarchical data structure (Rohrer et al., 2020). Considering the relatively small number of participating

Table 1
Conceptual knowledge and representational competence tests.

Knowledge type	ω AR	Control	Number of items	Item sample	Content covered
Prior representational competence	0.79	0.72	10	The figures show two field vectors of a vector field at the locations P and Q. Draw a corresponding vector field in field line representation directly in the box.	The test aims to assess the understanding and interpretation of electric or magnetic field representations, as well as the translation between field lines and vector field diagrams, and the recognition of field strength through visual cues, with a focus on graphical literacy in working with abstract physical representations.
Subsequent representational competence	0.78	0.75			
Prior conceptual knowledge	0.52	0.58		The figure below shows 2 bar magnets. The point P is located precisely in the center plane between the two magnets. Draw the field vector of the resulting magnetic field of both magnets at point P.	The test aims to evaluate the core competencies in magnetism at the high school level, such as understanding and drawing magnetic field vectors from bar magnets, horseshoe magnets, and current-carrying conductors, as well as the superposition of magnetic fields from multiple sources (e.g., two wires or magnets) and distinguishing correct from incorrect conceptual statements about magnetic poles, magnetic materials, and field interactions.
Subsequent conceptual knowledge	0.77	0.71		The figure below shows a horseshoe magnet lying flat on the table. The conductor swing is directed horizontally, running between the legs of the magnet parallel to its field lines. As soon as a current flows in the conductor, the magnetic fields of the horseshoe magnet and the conductor overlap. a. What does the resulting magnetic field look like? Describe and sketch your idea. b. Explain whether and in which direction a Lorentz force acts. 	The test evaluates students' ability to determine the direction of magnetic fields created by current-carrying conductors using the right-hand rule. Students are expected to predict the direction of the Lorentz force based on the interaction between the current and magnetic field vectors. The test also assesses the interpretation and creation of both field vector and field line representations. Additionally, it examines the ability to reason through superpositions of magnetic fields from multiple sources.

classes, we considered this approach more appropriate than modelling class as a random effect. To estimate the effect of the group, we included group membership as a dummy-coded variable (0 for the Control group and 1 for AR).

To address the second research question, Pearson correlation analyses were conducted to examine the relationship between conceptual knowledge and representational competence at both prior to the learning sequence (T1) and after it (T2).

We specified a cross-lagged path model to explore the predictive relationship between prior representational competence and conceptual knowledge, as well as the relationship between representational competence and conceptual knowledge outcomes. As a type of structural equation model, the cross-lagged path model examines the reciprocal relationships between variables measured at several time points (T1 and T2), which helps determine which variable predicts changes in the other variable over time.

In line with Newman (2014), we treated missing values using the full information maximum likelihood (FIML) method. For item-level scores, missing responses were coded as zero, reflecting incorrect answers. To examine whether the associations between representational competence and conceptual knowledge differed between groups, we used a multigroup cross-lagged path model, treating the "group" variable (AR vs. control) as a categorical grouping factor using the lavaan package in R (Rosseel, 2012). We specified a model in which all structural paths, intercepts, and residual variances were freely estimated across groups (i.e., the fully unconstrained model) and compared it to a model where the key cross-lagged path from prior representational competence to conceptual knowledge was constrained to be equal across groups. Due to the use of cluster-robust standard errors to account for the nested structure of the data (students nested within classrooms), formal chi-squared difference testing was not applicable. However, model fit indices (CFI, RMSEA, AIC, BIC) were identical across models, suggesting that constraining the path did not worsen model fit. The a priori power analysis indicated that, with a sample size of $N = 90$ is required to achieve a power of ≥ 0.80 for detecting a cross-lagged effect of $\beta = 0.25$ at $\alpha = .05$. Given that the per-group sample sizes were relatively modest ($n = 126$ and $n = 77$), results from the separate group analyses in the cross-lagged path model were interpreted primarily through confidence intervals to account for estimation uncertainty.

4. Results

4.1. Descriptive statistics

Descriptive statistics and Pearson correlations for prior and subsequent conceptual knowledge and representational competence are presented in Table S2 in the supplementary material, separately for the AR and control groups. Independent samples *t* tests revealed that students in the AR group outperformed those in the control group on all measures, both before and after the intervention.

At baseline, the AR group had significantly higher prior representational competence ($M = 0.55$, $SD = 0.19$) than the control group ($M = 0.43$, $SD = 0.16$), $t(182.21) = -5.03$, $p < .001$, $d = 0.67$, 95% CI [0.38, 0.96]. They also showed higher prior conceptual knowledge ($M = 0.47$, $SD = 0.13$) compared to the control group ($M = 0.42$, $SD = 0.12$), $t(174.66) = -2.80$, $p = .006$, $d = 0.39$, 95% CI [0.10, 0.67].

These group differences widened following the intervention. Post-intervention, the AR group scored significantly higher on representational competence ($M = 0.57$, $SD = 0.19$) than the control group ($M = 0.44$, $SD = 0.17$), $t(178.12) = -5.33$, $p < .001$, $d = 0.71$, 95% CI [0.42, 1.01]; and on conceptual knowledge ($M = 0.32$, $SD = 0.15$) compared to the control group ($M = 0.20$, $SD = 0.12$), $t(193.59) = -6.25$, $p < .001$, $d = 0.84$, 95% CI [0.54, 1.13].

All measured variables were significantly and positively correlated ($p < .001$), indicating moderate to strong linear relationships. In the AR group, prior representational knowledge was strongly associated with subsequent representational competence ($r = 0.80$). In contrast, prior representational competence showed a moderate positive correlation with conceptual knowledge after the intervention ($r = 0.46$). Likewise, prior conceptual knowledge was moderately linked to subsequent conceptual knowledge ($r = 0.56$). Prior representational competence correlated similarly with subsequent conceptual knowledge ($r = 0.55$).

Comparable patterns emerged in the control group ($n = 77$). Prior knowledge of representations was strongly correlated with later representational competence ($r = 0.82$), and prior conceptual knowledge showed a moderate association with subsequent conceptual understanding ($r = 0.63$).

Results regarding within-group improvement and between-group differences on the nine identical items from the pre- and post-tests of conceptual knowledge are reported in Supplementary Material Note 1.

4.2. Results for research question 1

The first research question aimed to examine how the AR and control groups differed in their conceptual knowledge after learning, while controlling for prior conceptual knowledge, representational competence, and the classroom session. Participants in the AR condition demonstrated significantly greater conceptual knowledge after the intervention compared to those in the control condition. The ANCOVA revealed a significant main effect of group, $F(1, 190) = 57.61$, $p < .001$, $\eta_p^2 = 0.23$, 95% CI [0.15, 1.00]. Students in the AR condition ($M = 0.32$) outperformed those in the control condition ($M = 0.20$), after adjusting for prior knowledge and classroom effects. All means reported are estimated marginal means computed via ANCOVA, controlling for prior representational competence, prior conceptual knowledge, and classroom session.

Additionally, prior conceptual knowledge was a strong predictor of post-test performance, $F(1, 190) = 115.09$, $p < .001$, $\eta_p^2 = 0.38$, 95% CI [0.29, 1.00], as was prior representational competence, $F(1, 190) = 22.81$, $p < .001$, $\eta_p^2 = 0.11$, 95% CI [0.05, 1.00]. A small but statistically significant effect of classroom membership was also observed, $F(9, 190) = 1.93$, $p = .045$, $\eta_p^2 = 0.08$, 95% CI [0.00, 1.00], suggesting that class-level factors explained some variance in the subsequent conceptual knowledge test performance. These findings suggest that learning in the AR environment promotes a deeper conceptual understanding, regardless of prior knowledge or class

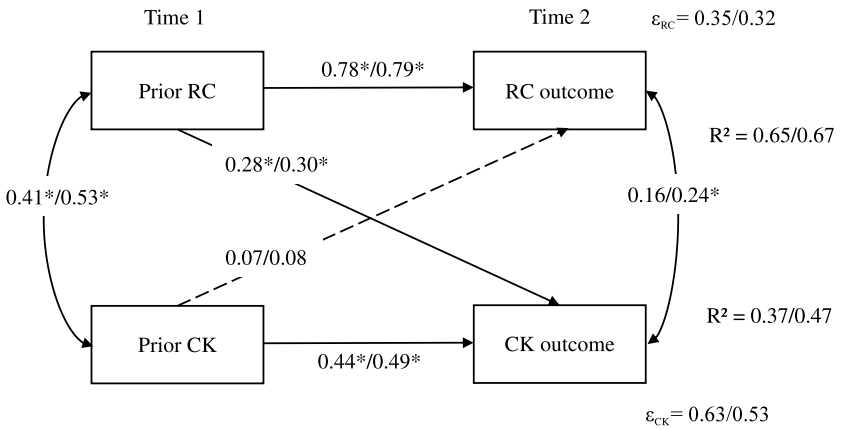


Fig. 5. Cross-lagged path model of prior conceptual knowledge (CK) and representational competence (RC), and conceptual knowledge and representational competence outcomes with standardized estimates for AR group and Control group
Note: Conceptual knowledge (CK); representational competence (RC); $N = 203$. $p < .01$. Solid lines represent statistically significant relationships; dashed lines represent non-significant.

allocation.

4.3. Results for research question 2

To address the second research question, the relationship between the conceptual knowledge and representational competence was examined at both prior (T1) and after the learning sequence (T2). The correlation between representational and conceptual knowledge prior to the intervention was moderate in both groups (AR: $r = 0.41$; Control: $r = 0.53$), which reflects a consistent relationship between these constructs prior to instruction. After the learning sequence, the relationship between representational competence and conceptual knowledge remained positive and of comparable magnitude (AR: $r = 0.47$; Control: $r = 0.58$). Contrary to H2.2, these results do not suggest an increase in the strength of the correlation from T1 to T2.

4.4. Results for research question 3

To address the third research question - namely, to what extent prior conceptual knowledge (CK_{t1}) and prior representational competence (RC_{t1}) at T1 (pre-learning) predict subsequent conceptual knowledge (CK_{t2}) and representational competence (RC_{t2}) at T2 - a series of cross-lagged path models was estimated. These models tested both autoregressive and cross-lagged paths to explore the reciprocal influence between conceptual knowledge and representational competence over time. Analyses were conducted separately for students in the AR-supported and traditional (Control) learning environments, and a multigroup comparison was used to evaluate whether these predictive relationships differed by instructional context. An overview of the main findings is shown in Fig. 5. The graph presents results for the AR and Control groups, separated by a slash (AR/Control) for ease of comparison.

As seen in Fig. 5, in the AR group ($n = 126$), both autoregressive paths were significant: CK_{t2} was predicted by CK_{t1} (95% CI [0.37, 0.67]), and RC_{t2} was predicted by RC_{t1} (95% CI [0.72, 0.87]). Additionally, RC_{t1} significantly predicted CK_{t2} (95% CI [0.15, 0.30]), indicating a meaningful cross-lagged influence from representational competence to subsequent conceptual knowledge. In contrast, the reverse cross-lagged path - from CK_{t1} to RC_{t2} - was not significant (95% CI [–0.10, 0.30]). The control group ($n = 77$) showed a similar pattern: CK_{t2} was significantly predicted by CK_{t1} (95% CI [0.36, 0.60]), and RC_{t2} was predicted by RC_{t1} (95% CI [0.76, 0.89]). Again, RC_{t1} significantly predicted CK_{t2} (95% CI [0.10, 0.34]), while CK_{t1} did not significantly predict RC_{t2} (95% CI [–0.23, 0.41]).

To assess whether the cross-lagged paths differed significantly across groups, we estimated a multigroup model constraining the $RC_{t1} \rightarrow CK_{t2}$ path to be equal in both groups, and compared it to an unconstrained model. A chi-square difference test is not applicable due to the use of cluster-robust standard errors. However, fit indices (AIC, BIC, CFI, RMSEA) were identical across the constrained and unconstrained models, indicating that the cross-lagged path from RC_{t1} to CK_{t2} did not differ significantly between groups. The path estimates visible in Fig. 5, together with their confidence intervals, also indicate that the parameters were very similar in both conditions. These results suggest that representational competence consistently predicts later conceptual knowledge regardless of instructional format, though other model parameters may still vary by condition.

5. Discussion

5.1. General discussion and implications

The aim of the present study was to explore how prior representational competence and conceptual knowledge relate to the subsequent development of conceptual knowledge and representational competence when learning in an AR-supported environment. To examine this, we investigated how prior conceptual knowledge and representational competence affect subsequent conceptual knowledge and representational competence during the acquisition of conceptual understanding of the Lorentz force, comparing an AR-enhanced experimental setup with a traditional setup. We also looked into the difference in subsequent conceptual knowledge between the AR and control groups.

We hypothesized (H1) that the AR group would demonstrate significantly higher conceptual knowledge after the intervention compared to the control group. This hypothesis was confirmed. We also hypothesized a positive relationship between conceptual knowledge and representational competence of moderate to large magnitude (a linear correlation between 0.30 and 0.70), both before and after the learning sequence (H2.1). This hypothesis was supported. We also expected the correlation between the two constructs to be stronger after learning (at T2) than before learning (at T1), with an increase in correlation of about 20% (e.g., from 0.45 to 0.54). This hypothesis was not confirmed. We expected (H3.1) a positive relationship between conceptual knowledge at T2 (post-intervention) and representational competence at T1 (pre-intervention), controlling for prior conceptual knowledge, in both the traditional and AR-supported learning environments. This hypothesis was supported. However, contrary to our hypothesis H3.2, the relationship did not significantly differ across the conditions. Regarding the relationship between prior conceptual knowledge and subsequent representational competence (H3.3), we hypothesized that prior conceptual knowledge would be a positive predictor of subsequent representational competence when controlling for prior representational competence. This hypothesis was not confirmed.

5.1.1. Discussion for research question 1

Our findings revealed that the AR group outperformed the control group in subsequent conceptual knowledge. This supports previously-gained evidence suggesting that AR visualizations of invisible phenomena can positively impact learning outcomes (Cai et al., 2017; Ibáñez et al., 2014). In particular, in domains such as electromagnetism, where the subject matter involves intangible concepts like field lines and forces, AR has been shown to aid learners' conceptual understanding by overlaying dynamic visualizations

directly onto real-world contexts (Dunleavy and Dede, 2013; Yoon et al., 2012).

Our findings align with prior research that indicates that external representations can support learning by offloading internal processing demands (Ainsworth, 2006). Moreover, our findings are in line with studies suggesting that when AR is well-designed and meaningfully embedded within the instructional context, it can promote learning (Akçayır & Akçayır, 2017; Garzón et al., 2020). In other words, rather than fully compensating for the role of representational competence in conceptual understanding, AR appears to function as a scaffold that aids learners in engaging with complex representational content, without replacing the underlying representational competence required.

5.1.2. Discussion for research question 2

We identified moderate correlations between representational competence and conceptual knowledge both before and after the learning sequence. This supports H2.1. and aligns with prior research (Edelsbrunner et al., 2023; Nitz et al., 2014), which has consistently shown positive associations between these constructs. These findings may suggest a stable and meaningful link between representational and conceptual understanding across time and context. At the same time, H2.2 was not supported, as the correlation between conceptual knowledge and representational competence did not increase from T1 to T2. The absence of an increase in the strength of the relationship may be due to the learning sequence's primary focus on building conceptual knowledge. That may have led to an indirect improvement in representational competence, resulting in parallel rather than integrated development of the two constructs.

5.1.3. Discussion for research question 3

As for the cross-lagged effects, we found a significant relationship between prior representational competence and subsequent conceptual knowledge in both groups. This relationship was of comparable magnitude in the control group (traditional experimental setup) and the AR group (setup with integrated representations). These findings support the idea that representational competence functions as an important precondition for conceptual learning in STEM contexts, as it enables learners to decode, interpret, and make sense of domain-specific representations central to understanding complex scientific phenomena. This aligns with previous results reported by Edelsbrunner et al. (2023), Nitz et al. (2014), and Crowder et al. (2025). Our findings extend this line of research by highlighting the importance of representational competence for conceptual knowledge development in AR-supported environments as well. Notably, the comparable strength of predictive paths across conditions suggests that AR did not fundamentally alter the necessity of integrating representational competence with conceptual knowledge for conceptual learning. Instead, it appears to have operated within the same cognitive architecture, enhancing the accessibility of key representations, but not eliminating the necessity of representational thinking. This nuanced result resonates with findings by Altmeyer et al. (2024). They argue that AR supports learners most effectively when it complements, existing instructional strategies. It is also in line with prior evidence on the compensatory effects that visualizing representations can have on knowledge acquisition (Ainsworth, 2006; Kohnle & Passante, 2017).

In contrast, contrary to our hypothesis, we found no predictive relationship from prior conceptual knowledge to subsequent representational competence. One possible interpretation of this finding is that learners did not draw on their available topic-specific conceptual knowledge while developing representational competence. Yet, one has to put in context that our measure of representational competence acts topic-independently. Consequently, it may be the case that learners have built up representational competence but have not yet managed to transfer it from the topic-specific representations that they worked with during experimentation to the more topic- (or domain-) general representations in the assessment. Individual differences in representational competence were extremely stable from pre-to posttest, as indicated by the large regression weight, and there was no average change in learners' representational competence from pre-to posttest, so either learners' representational competence did not change during the experimentation sessions, or our measure was not sensitive to changes that took place. Given that representational competence at pretest was an apt predictor of the amount of conceptual knowledge built up during the intervention, it appears that our measure was sensitive to the representational competence the learners drew upon in the experimental setup. Overall, we cannot determine whether the learners did not implicitly acquire representational competence during the session, our measure was not sensitive towards such a change, or whether conceptual knowledge was indeed not a factor affecting change. Future research is required to disentangle these possibilities and further test the view that developing representational competence lays the foundation for conceptual learning, rather than emerging from it (Edelsbrunner and Hofer, 2024).

5.2. Limitations and future research

Several limitations of the present study should be addressed in future research. Firstly, the students in the sample were Gymnasium students (the highest level of secondary education in the German system). This means that, academically, the students show above-average performance (Becker et al., 2012; Köller & Baumert, 2001). Moreover, due to implementation constraints, randomization on an individual level and collecting data simultaneously for both groups were not feasible. Class membership and prior knowledge as a strong general indicator of student expertise were controlled for in the statistical analyses (Stallasch et al., 2024), and various relevant variables (i.e., age, grade, school type) did not differ significantly between the groups. Still, future research would benefit from replicating the study with a more heterogeneous sample, including different school types and age groups. This consideration also relates to the power aspect of the study. Even though the a priori power analysis was conducted, it did not account for classroom-level randomization. To gain a more generic understanding and a broader range of performance patterns, future research should examine a broader population with more diverse academic backgrounds.

There are also certain limitations associated with the instructional design employed. Before the pre-tests, the learners were shown a

video aimed at equipping them with the baseline understanding of the concepts and specifically representations necessary to proceed to the intervention. The representations depicted in the video were similar to those in the representational prior knowledge test. While on the one hand, watching the videos may have activated the learners′ prior knowledge, on the other hand, showing them immediately before the test may have influenced the outcome of the prior representational competence test. To assess prior knowledge of representations more accurately, it would be preferable to show the videos after administering the prior knowledge test.

Certain limitations are also related to the use of AR in this study. As its novelty can introduce usability challenges (Akçayır & Akçayır, 2017), learners initially needed more time to orient themselves and understand how to interact effectively with the AR system, which has led to certain disruptions and delays. For this reason, the total study duration sometimes exceeded 4 h, which may have led to fatigue and affected the post-test performance. Some students also reported high levels of exhaustion and feeling overwhelmed. Future research should consider reducing the study duration and including the measurement of affective variables, i.e., interest and engagement, to control for their effect in the analyses.

At the same time, considering the nuanced study design, we believe that both our findings and instructional design insights can inform instructional decisions regarding the effective integration of AR in real classroom settings. Specifically, our results suggest that AR should be accompanied by explicit scaffolds such as clear prior knowledge activation tasks and pre-teaching activities to activate relevant representations. Moreover, the use of AR should be well-justified from the didactical perspective and embedded within coherent and cohesive lesson sequences.

To test the generalizability of our findings, future research could examine the relationship between representational competence and conceptual knowledge in other STEM domains (e.g., biology) while maintaining the same key affordance - visualizing otherwise invisible objects. Given the framework proposed by Kozlova et al. (2025), which links environmental affordances to specific learning outcomes, similar effects might be expected in topics where this affordance is pedagogically relevant. For instance, dynamic representations in cellular biology could similarly support the development of conceptual understanding. However, whether such effects replicate may depend not only on the type of affordance but also on the instructional design, particularly how well the visualizations are scaffolded. Understanding the nature of the relationship between representational competence and conceptual knowledge more deeply could help identify the most effective instructional designs for supporting learners with weaker representational competence in grasping complex scientific phenomena.

For this reason, future AR research should focus on exploring the role of representational competence in conceptual knowledge formation across a wider range of STEM contexts and on investigating the long-term effects of AR use in real classroom environments. The collection of data, such as eye-tracking measures (e.g., fixation duration, gaze transitions), could provide more fine-grained insights into learners' behavior and attentional engagement during AR-supported learning. It could also manipulate the degree of representational support in AR more directly or examine learners with a wider range of representational abilities to clarify these effects. Such work could contribute to the more targeted development of AR technologies in STEM education, particularly for topics where external representations are essential (e.g., electromagnetism). To better grasp which instructional strategies in AR most effectively enhance learning, future studies should examine how different design decisions within AR - such as varying levels of support or integration - may interact with the learners ' prior conceptual knowledge and representational competence.

6. Conclusion

This study explored how prior conceptual knowledge and representational competence predict learning outcomes in AR-supported versus traditional learning environments. Across both conditions, representational competence significantly predicted conceptual knowledge acquisition, highlighting its central role in STEM learning. Augmented reality did not fundamentally eliminate the need to integrate representational and conceptual knowledge; rather, it appears to have supported this process by enhancing the accessibility and salience of key representations. In this way, AR functioned within the same cognitive architecture, facilitating - but not replacing - the representational competence necessary for the conceptual knowledge acquisition.

CRediT authorship contribution statement

Kozlova Zoya: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Edelsbrunner Peter A.:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Warkentin Max:** Software, Project administration, Data curation. **Hoyer Christoph:** Writing – review & editing, Software, Resources, Project administration, Data curation. **Küchemann Stefan:** Writing – review & editing, Conceptualization. **Kuhn Jochen:** Writing – review & editing, Supervision, Resources, Conceptualization. **Hofer Sarah I.:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors confirm that there are no competing financial or non-financial interests that could be perceived to influence the outcomes of this study. No financial support, consulting relationships, or personal connections exist that could impact the work presented here. The research was conducted independently of any commercial or organizational influence.

Acknowledgements

This project was funded by DFG within the project CESAR (Project number 471917560) and by LMUexcellent, funded by the Federal Ministry of Education and Research (BMBF) and the Free State of Bavaria under the Excellence Strategy of the Federal Government and the Länder.

We would like to thank Raphael Cera, Denise Zhu, Tobias Schnaitmann and Anja Müller for their support during the data collection phase.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compedu.2026.105658.

Data availability

The dataset can be accessed at https://osf.io/download/687781b39a2e720860feb5cd/.

References

Ainsworth, S. (2006). DeFT: A conceptual framework for considering learning with multiple representations. *Learning and instruction, 16*(3), 183–198.

Akçayır, M., & Akçayır, G. (2017). Advantages and challenges associated with augmented reality for education: A systematic review of the literature. *Educational Research Review, 20*, 1–11. https://doi.org/10.1016/j.edurev.2016.11.002

Alexander, P. A., & Judy, J. E. (1988). The interaction of domain-specific and strategic knowledge in academic performance. *Review of Educational Research, 58*(4), 375–404. https://doi.org/10.2307/1170279

Alexander, P. A., & Murphy, P. K. (1999). Nurturing the seeds of transfer: A domain-specific perspective. *International Journal of Educational Research, 31*(7), 561–576. https://doi.org/10.1016/S0883-0355(99)00024-5

Altmeyer, K., Brünken, R., Kuhn, J., & Malone, S. (2024). The role of cognitive learner prerequisites for cognitive load and learning outcomes in AR-supported lab work. *Education Sciences, 14*(11), 1161.

Altmeyer, K., Kapp, S., Thees, M., Malone, S., Kuhn, J., & Brünken, R. (2020). The use of augmented reality to foster conceptual knowledge acquisition in STEM laboratory courses—Theoretical background and empirical results. *British Journal of Educational Technology, 51*(3), 611–628. https://doi.org/10.1111/bjet.12900

Amores-Valencia, A., Burgos, D., & Branch-Bedoya, J. W. (2023). The influence of augmented reality (AR) on the motivation of high school students. *Electronics, 12*(22), 4715. https://doi.org/10.3389/fpsyg.2022.1011409

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of the mind. *Psychological Review, 111*(4), 1036–1060. https://doi.org/10.1037/0033-295X.111.4.1036

Azuma, R. T. (1997). A survey of augmented reality. *Presence: Teleoperators and Virtual Environments, 6*(4), 355–385. https://doi.org/10.1162/pres.1997.6.4.355

Baxter, G. P., Elder, A. D., & Glaser, R. (1996). Knowledge-based cognition and performance assessment in the science classroom. *Educational Psychologist, 31*(2), 133–140.

Becker, M., Lüdtke, O., Trautwein, U., Köller, O., & Baumert, J. (2012). The differential effects of school tracking on psychometric intelligence: Do academic-track schools make students smarter? *Journal of Educational Psychology, 104*(3), 682–699. https://doi.org/10.1037/a0027608

Bhagat, K. K., Yang, F. Y., Cheng, C. H., Zhang, Y., & Liou, W. K. (2021). Tracking the process and motivation of math learning with augmented reality. *Educational Technology Research and Development, 69*(6), 3153–3178. https://doi.org/10.1007/s11423-021-10066-9

Billinghurst, M., Clark, A., & Lee, G. (2015). A survey of augmented reality. *Foundations and Trends® in Human–Computer Interaction, 8*(2–3), 73–272. https://doi.org/10.1561/1100000049

Binder, T., Sandmann, A., Sures, B., Friege, G., Theyssen, H., & Schmiemann, P. (2019). Assessing prior knowledge types as predictors of academic achievement in the introductory phase of biology and physics study programs using logistic regression. *International Journal of STEM Education, 6*(1). https://doi.org/10.1186/s40594-019-0189-9. Article 33.

Bittermann, A., McNamara, D., Simonsmeier, B. A., & Schneider, M. (2023). The landscape of research on prior knowledge and learning: A bibliometric analysis. *Educational Psychology Review*. https://doi.org/10.1007/s10648-023-09775-9

Buchner, J., Buntins, K., & Kerres, M. (2022). The impact of augmented reality on cognitive load and performance: A systematic review. *Journal of Computer Assisted Learning, 38*(1), 285–303. https://doi.org/10.1111/jcal.12553

Buchner, J., & Kerres, M. (2023). Media comparison studies dominate comparative research on augmented reality in education. *Computers & Education, 195*, Article 104711. https://doi.org/10.1016/j.compedu.2022.104711

Bujak, K. R., Radu, I., Catrambone, R., MacIntyre, B., Zheng, R., & Golubski, G. (2013). A psychological perspective on augmented reality in the mathematics classroom. *Computers & Education, 68*, 536–544. https://doi.org/10.1016/j.compedu.2013.02.017

Cai, S., Chiang, F. K., Sun, Y., Lin, C., & Lee, J. J. (2017). Applications of augmented reality-based natural interactive learning in magnetic field instruction. *Interactive Learning Environments, 25*(6), 778–791. https://doi.org/10.1080/10494820.2016.1181094

Carey, S. (2000). Science education as conceptual change. *Journal of Applied Developmental Psychology, 21*(1), 13–19. https://doi.org/10.1016/S0193-3973(99)00046-5

Coban, A., Dzsotjan, D., Küchemann, S., et al. (2025). AI support meets AR visualization for Alice and Bob: Personalized learning based on individual ChatGPT feedback in an AR quantum cryptography experiment for physics lab courses. *EPJ Quantum Technol., 12*, 15. https://doi.org/10.1140/epjqt/s40507-025-00310-z

Crowder, C. J., Ward, L. W., Popova, M., Komperda, R., Rotich, F., & Raker, J. R. (2025). Testing a reciprocal causation model between the organic chemistry representational competence assessment and examination performance in postsecondary organic chemistry. *Journal of Chemical Education*. https://doi.org/10.1021/acs.jchemed.5c00169

Dunleavy, M., & Dede, C. (2013). Augmented reality teaching and learning. *Handbook of research on educational communications and technology*, 735–745.

Edelsbrunner, P. A., & Hofer, S. I. (2024). Examining and comparing the relation between representational competence and conceptual knowledge across four samples. In *Frontiers in education, 9*Frontiers Media SA, Article 1459603. https://doi.org/10.3389/feduc.2024.1459603.

Edelsbrunner, P. A., Malone, S., Hofer, S. I., Küchemann, S., Kuhn, J., Schmid, R., … Lichtenberger, A. (2023). The relation of representational competence and conceptual knowledge in female and male undergraduates. *International Journal of STEM Education, 10*(1), 44. https://doi.org/10.1186/s40594-023-00435-6

Edelsbrunner, P. A., Simonsmeier, B. A., & Schneider, M. (2025). The Cronbach's alpha of domain-specific knowledge tests before and after learning: A meta-analysis of published studies. *Educational Psychology Review, 37*(1), 4. https://doi.org/10.31234/osf.io/m8d7t

Engelbrecht, J., Harding, A., & Potgieter, M. (2005). Undergraduate students' performance and confidence in procedural and conceptual mathematics. *International Journal of Mathematical Education in Science and Technology, 36*(7), 701–712. https://doi.org/10.1080/00207390500271107

Garzón, J., Baldiris, S., Gutiérrez, J., & Pavón, J. (2020). How do pedagogical approaches affect the impact of augmented reality on education? A meta-analysis and research synthesis. *Educational Research Review, 31*, 100334.
Gilbert, J. K. (2008). Visualization: An emergent field of practice and enquiry in science education. In *Visualization: Theory and practice in science education* (pp. 3–24). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-1-4020-5267-5_1.
Gonnermann-Müller, J., & Krüger, J. M. (2025). Unlocking augmented reality learning design based on evidence from empirical cognitive load studies—A systematic literature review. *Journal of Computer Assisted Learning, 41*(1), Article e13095. https://doi.org/10.1111/jcal.13095
Hailikari, T., Nevgi, A., & Komulainen, E. (2007). Academic self-beliefs and prior knowledge as predictors of student achievement in mathematics: A structural model. *Educational Psychology, 28*(1), 59–71. https://doi.org/10.1080/01443410701413753
Hsu, Y. S., Lin, Y. H., & Yang, B. (2017). Impact of augmented reality lessons on students' STEM interest. *Research and Practice in Technology Enhanced Learning, 12*, 1–14. https://doi.org/10.1186/s41039-016-0039-z
Ibáñez, M. B., & Delgado-Kloos, C. (2018). Augmented reality for STEM learning: A systematic review. *Computers & Education, 123*, 109–123. https://doi.org/10.1016/j.compedu.2018.05.002
Ibáñez, M. B., Di Serio, Á., Villarán, D., & Kloos, C. D. (2014). Experimenting with electromagnetism using augmented reality: Impact on flow student experience and educational effectiveness. *Computers & Education, 71*, 1–13. https://doi.org/10.1016/j.compedu.2013.09.004
Jiang, Y., Wang, Y., & Wang, Y. (2024). Virtual reality and augmented reality-supported K–12 STEM learning: A systematic review. *Education and Information Technologies, 29*, 123–145. https://doi.org/10.1007/s10639-024-13210-z
Kendeou, P., & Van Den Broek, P. (2007). The effects of prior knowledge and text structure on comprehension processes during reading of scientific texts. *Memory & Cognition, 35*(7), 1567–1577. https://doi.org/10.3758/BF03193491
Kohnle, A., & Passante, G. (2017). Characterizing representational learning: A combined simulation and tutorial on perturbation theory. *Physical Review Physics Education Research, 13*, Article 20131. https://doi.org/10.1103/PhysRevPhysEducRes.13.020131
Köller, O., & Baumert, J. (2001). Leistungsgruppierungen in der Sekundarstufe I: Ihre Konsequenzen für die Mathematikleistung und das mathematische Selbstkonzept der Begabung. *Zeitschrift für Padagogische Psychologie, 15*(2), 99–110.
Kozlova, Z., Bach, K. M., Edelsbrunner, P. A., & Hofer, S. I. (2025). Bringing learners into focus: A systematic review of learner characteristics in AR-supported STEM education. *Learning and Individual Differences, 122*, Article 102727. https://doi.org/10.1016/j.lindif.2025.102727
Kozma, R., Chin, E., Russell, J., & Marx, N. (2000). The roles of representations and tools in the chemistry laboratory and their implications for chemistry learning. *The Journal of the Learning Sciences, 9*(2), 105–143. https://doi.org/10.1207/s15327809jls0902_1
Küchemann, S., Malone, S., Edelsbrunner, P., Lichtenberger, A., Stern, E., Schumacher, R., … Kuhn, J. (2021). Inventory for the assessment of representational competence of vector fields. *Physical Review Physics Education Research, 17*(2), Article 020126. https://doi.org/10.1103/PhysRevPhysEducRes.17.020126
Lawson, A. P., Martella, A. M., LaBonte, K., Delgado, C. Y., Zhao, F., Gluck, J. A., Munns, M. E., Wells LeRoy, A., & Mayer, R. E. (2024). Confounded or controlled? A systematic review of media comparison studies involving immersive virtual reality for STEM education. *Educational Psychology Review, 36*(3), 69. https://doi.org/10.1007/s10648-024-09908-8
Lin, H. C. K., Chen, M. C., & Chang, C. K. (2015). Assessing the effectiveness of learning solid geometry by using an augmented reality-assisted learning system. *Interactive Learning Environments, 23*(6), 799–810. https://doi.org/10.1080/10494820.2013.817435
Ling, Y., Zhu, P., & Yu, J. (2021). Which types of learners are suitable for augmented reality? A fuzzy set analysis of learning outcomes configurations from the perspective of individual differences. *Educational Technology Research and Development, 69*(6), 2985–3008.
Moncur, B., Galvez Trigo, M. J., & Mortara, L. (2023). Augmented reality to reduce cognitive load in operational decision-making. In D. D. Schmorrow, & C. M. Fidopiastis (Eds.), *Augmented cognition. HCII 2023. Lecture notes in computer science, 14019*. https://doi.org/10.1007/978-3-031-35017-7_21. Springer, Cham.
Newman, D. A. (2014). Missing data: Five practical guidelines. *Organizational Research Methods, 17*(4), 372–411. https://doi.org/10.1177/1094428114548590
Nitz, S., Ainsworth, S. E., Nerdel, C., & Prechtl, H. (2014). Do student perceptions of teaching predict the development of representational competence and biological knowledge? *Learning and Instruction, 31*, 13–22. https://doi.org/10.1016/j.learninstruc.2013.12.003
Rau, M. A. (2017). Conditions for the effectiveness of multiple visual representations in enhancing STEM learning. *Educational Psychology Review, 29*, 717–761. https://doi.org/10.1007/s10648-016-9365-3
Rau, M. A. (2018). Making connections among multiple visual representations: How do sense-making skills and perceptual fluency relate to learning of chemistry knowledge? *Instructional Science, 46*(2), 209–243. https://doi.org/10.1007/s11251-017-9431-3
Rohrer, D., Dedrick, R. F., Hartwig, M. K., & Cheung, C. N. (2020). A randomized controlled trial of interleaved mathematics practice. *Journal of Educational Psychology, 112*(1), 40. https://doi.org/10.1037/edu0000367
Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software, 48*(2), 1–36. http://www.jstatsoft.org/v48/i02/.
R Core Team. (2026). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals
Sahin, D., & Yilmaz, R. M. (2020). The effect of augmented reality technology on middle school students' achievements and attitudes towards science education. *Computers & Education, 144*, Article 103710. https://doi.org/10.1016/j.compedu.2019.103710
Sarwar, G. S., & Trumpower, D. L. (2015). Effects of conceptual, procedural, and declarative reflection on students' structural knowledge in physics. *Educational Technology Research and Development, 63*, 185–201. https://doi.org/10.1007/s11423-015-9375-y
Scheid, J., Müller, A., Hettmannsperger, R., & Schnotz, W. (2018). Representational competence in science education: From theory to assessment. In *Towards a framework for representational competence in science education* (pp. 263–277). https://doi.org/10.1007/978-3-319-89945-9_13
Schneider, M., & Simonsmeier, B. A. (2025). How does prior knowledge affect learning? A review of 16 mechanisms and a framework for future research. *Learning and Individual Differences, 122*, Article 102744. https://doi.org/10.1016/j.lindif.2025.102744
Simonsmeier, B. A., Flaig, M., Deiglmayr, A., Schalk, L., & Schneider, M. (2022). Domain-specific prior knowledge and learning: A meta-analysis. *Educational Psychologist, 57*(1), 31–54. https://doi.org/10.1080/00461520.2021.1939700
Sırakaya, M., & Alsancak Sırakaya, D. (2022). Augmented reality in STEM education: A systematic review. *Interactive Learning Environments, 30*(8), 1556–1569. https://doi.org/10.1080/10494820.2020.1722713
Stadler, M., Sailer, M., & Fischer, F. (2021). Knowledge as a formative construct: A good alpha is not always better. *New Ideas in Psychology, 60*, Article 100832. https://doi.org/10.1016/j.newideapsych.2020.100832
Stallasch, S. E., Lüdtke, O., Artelt, C., Hedges, L. V., & Brunner, M. (2024). Single-and multilevel perspectives on covariate selection in randomized intervention studies on student achievement. *Educational Psychology Review, 36*(4), 112. https://doi.org/10.1007/s10648-024-09898-7
Steffen, J. H., Gaskin, J. E., Meservy, T. O., Jenkins, J. L., & Wolman, I. (2019). Framework of affordances for virtual reality and augmented reality. *Journal of Management Information Systems, 36*(3), 683–729. https://doi.org/10.1080/07421222.2019.1628877
Stieff, M. (2011). Improving representational competence using molecular simulations embedded in inquiry activities. *Journal of Research in Science Teaching, 48*(10), 1137–1158. https://doi.org/10.1002/tea.20438
Sweller, J. (2024). Cognitive load theory and individual differences. *Learning and Individual Differences, 110*, Article 102423.
Taber, K. S. (2018). The use of Cronbach's alpha when developing and reporting research instruments in science education. *Research in Science Education, 48*(6), 1273–1296. https://doi.org/10.1007/s11165-016-9602-2
Thees, M., Kapp, S., Strzys, M. P., Beil, F., Lukowicz, P., & Kuhn, J. (2020). Effects of augmented reality on learning and cognitive load in university physics laboratory courses. *Computers in Human Behavior, 108*, Article 106316. https://doi.org/10.1016/j.chb.2020.10631
Warkentin, M., Altmeyer, K., Liang, Y., Steinmacher, B., Gränz, B., Lichtenberger, A., Küchemann, S., Kuhn, J., & Hoyer, C. (2025). Hands-on experiment supported by augmented reality smartglasses for learning the Lorentz force. *arXiv preprint* https://arxiv.org/abs/2502.06421.
Witherby, A. E., & Carpenter, S. K. (2022). The rich-get-richer effect: Prior knowledge predicts new learning of domain-relevant information. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 48*(4), 483–498. https://doi.org/10.1037/xlm0000996

Wu, H. K., Lee, S. W. Y., Chang, H. Y., & Liang, J. C. (2013). Current status, opportunities, and challenges of augmented reality in education. *Computers & Education, 62*, 41–49. https://doi.org/10.1016/j.compedu.2012.10.024

Yenilmez, A., Sungur, S., & Tekkaya, C. (2006). Students' achievement in relation to reasoning ability, prior knowledge, and gender. *Research in Science & Technological Education, 24*(1), 129–138. https://doi.org/10.1080/02635140500485334

Yoon, S. A., Elinich, K., Wang, J., et al. (2012). Using augmented reality and knowledge-building scaffolds to improve learning in a science museum. *Computer Supported Learning, 7*, 519–541. https://doi.org/10.1007/s11412-012-9156-x

Yu, S., Liu, Q., Ma, J., Le, H., & Ba, S. (2022). Applying augmented reality to enhance physics laboratory experience: Does learning anxiety matter? *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2022.2057547

Zumbach, J., von Kotzebue, L., & Pirklbauer, C. (2022). Does augmented reality also augment knowledge acquisition? *Journal of Educational Computing Research, 60* (5), 1325–1346. https://doi.org/10.1177/07356331211062945

Zuo, R., Wenling, L., & Xuemei, Z. (2025). Augmented reality and student motivation: A systematic review (2013-2024). *Journal of Computers for Science and Mathematics Learning, 2*(1), 38–52. https://doi.org/10.70232/jcsml.v2i1.23

Hidayat, R., & Wardat, Y. (2024). A systematic review of augmented reality in science, technology, engineering and mathematics education. *Education and Information Technologies*, *29*(8), 9257-9282.Alexander, P.A. (1992). Domain knowledge: Evolving themes and emerging concerns. *Educational Psychologist*, *27*(1), 33-51. 10.1207/s15326985ep2701_4.